

Project Design Phase

Problem – Solution Fit Template

Date	24 February 2026
Team ID	LTVIP2026TMIDS61292
Project Name	Exploratory Analysis of Rainfall Data in India for Agriculture
Maximum Marks	2 Marks

1. Target Customer

Primary Customers: Farmers dependent on rainfall for crop cultivation. Secondary Customers: Agricultural planners and policymakers.

2. Core Problem Identified

Farmers face uncertainty due to unpredictable rainfall patterns. Lack of reliable short-term rainfall prediction leads to poor irrigation planning, incorrect fertilizer usage, crop damage, and financial loss.

3. Existing Situation

- Farmers rely on traditional knowledge or delayed weather forecasts.
- Manual decision-making increases risk during irregular climate conditions.
- No simple, localized prediction tool accessible via web interface.

4. Proposed Solution

The proposed solution is a Machine Learning-based Rainfall Prediction System that uses a Random Forest classification model (85.69% accuracy) deployed through a Flask web application. The system accepts weather parameters and provides real-time rainfall probability along with agricultural advisory.

5. Problem–Solution Fit Justification

- The system directly addresses rainfall uncertainty.
- Provides instant prediction instead of delayed forecasting.
- Offers actionable advisory recommendations (irrigation, fertilizer planning).
- Simple web interface ensures usability for non-technical users.

6. Behavioral Fit & Adoption Strategy

- Farmers already use mobile browsers → Web-based solution fits existing behavior.
- Simple input form reduces complexity.
- Advisory recommendations build trust through practical value.

- Frequent usage encouraged during planting and irrigation cycles.

7. Value Proposition

The Rainfall Prediction System reduces agricultural risk, improves decision-making, and enhances crop productivity by combining data-driven prediction with real-time advisory support.

Problem - Solution Fit: Rainfall Prediction System

